

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

V Semester of B. Sc. (Biotechnology/Microbiology) (Backlog) Examination, March 2018

Course code & Name: MI 501 & Principles of Environmental Biotechnology

Date: 29/03/2018 Day: Thursday Time: 1:30 P.M. To 1:50 P.M. Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. 10

1. Chemical hazards which interfere with the normal functioning of natural hormonal systems can be classified as 01
(i) Mutagens (ii) Endocrine disruptors
(iii) Teratogens (iv) Methanogens
2. Anaerobic digestion by gut microbes in digestive tract of the ruminant animals, and termite mounds is the natural source of which of the following greenhouse gas. 01
(i) Methane (ii) CFC
(iii) Ozone (iv) All of them
3. The following test is used to indicate the presence of coliform bacteria in water samples by detecting acid and gas production upon lactose fermentation. 01
(i) MPN (ii) BOD
(iii) COD (iv) TDS
4. Which of the following drinking water disinfection method does not generate disinfection by-products (DBPs). 01
(i) UV-irradiation (ii) Chlorination
(iii) Ozonation (iv) All of them

5. The use of living plants and their associated microorganisms for removal, degradation, or containment of harmful environmental pollutants is known as 01
- (i) Phytoremediation (ii) Biosorption
(iii) Bioleaching (iv) Biomagnification
6. In which of the following process microorganisms are utilized for the solubilization of base metals from low-grade ores. 01
- (i) Phytovolatilization (ii) Biodegradation
(iii) Bioleaching (iv) All of them
7. The following filamentous bacteria are responsible for the 'bulking' of sludge in the activated sludge system. 01
- (i) *Prochlorococcus* (ii) *Geotrichum*
(iii) *Cryptosporidium* (iv) *Sphaerotilus*
8. The following is a controlled microbial aerobic decomposition process with the formation of stabilized organic materials that may be used as soil conditioner. 01
- (i) Biosorption (ii) Composting
(iii) Bioleaching (iv) Anaerobic digestion
9. The following test measures the amount of dissolved oxygen consumed by microorganisms during the biochemical oxidation of organic matter. 01
- (i) COD (ii) BOD
(iii) TOC (iv) TSS
10. In the following air sampling method the airborne particles/bioaerosols are entrapped into a liquid matrix. 01
- (i) Impingement (ii) Impaction
(iii) Centrifugation (iv) All of them

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Course code & Name: MI 501 & Principles of Environmental Biotechnology

Date: 29/03/2018 Day: Thursday Time: 1:50 P.M. To 3:30 P.M. Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION – I

Q - II Answer the following questions as directed (answer any two) 10

1. Explain the Most Probable Number (MPN) test to detect total *Coliform* bacteria in water samples. 05
2. Describe in detail the primary and secondary processes for treatment of domestic wastewater. 05
3. Write a short note on composting of solid wastes. 05

SECTION – II

Q-III Answer the following questions as directed (answer any five) 15

1. Explain briefly different types of phytoremediation systems. 03
2. Write a short note on greenhouse effect. 03
3. Define the following terms: 03
(1) Bioaerosols (2) Biosorption (3) Bioleaching
4. Explain different methods for biological treatment of air pollution. 03
5. What is BOD? Briefly describe the procedure of the 5-day BOD test. 03
6. Describe the criteria for an ideal indicator microorganism to detect fecal contamination in water samples. 03
7. Define xenobiotics and biodegradation. Describe the possible outcomes of biodegradation. 03